GGG-H-33a
FEBRUARY 10, 1958
SUPERSEDING
Int. Fed. Spec GGG-H-0033 (NAVY-SHIPS)
20 December 1956,
Fed. Spec. GGG-H-86a (in part) and
Fed. Spec. LLL-M-71a (in part)
September 4, 1956
July 6, 1955
(See 6.3)

FEDERAL SPECIFICATION

HAMMERS, HAND AND MALLETS, SURFACE PROTECTIVE; HOLDERS AND ADAPTERS, INSERTED HAMMER FACE, AND FACES, HAMMER, INSERTED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers hammers, mallets, adapters, and inserted faces which are used to strike and provide protection to the surfaces of metals and finished products.

1.1.1 *Federal specification coverage.*—Federal specifications do not cover all types and sizes of the commodity indicated by the title of the specification, but are intended to cover only those most generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types, classes, styles, and designs.*—Hammers, mallets, adapters, and inserts shall be of the following types, classes, styles, and designs as specified (see 6.2):

Type I.—Hammer, inserted face.
Class 1.—Split head.
Style A.—Copper inserts.
Style B.—Rawhide inserts.
Style C.—Lead inserts.
Style D.—Plastic inserts.
Style E.—Rubber inserts.
Class 2.—Press-on head.
Style D.—Plastic inserts.
Design a.—Regular.
Design b.—No bounce.
Class 3.—Screw-in head.
Style D.—Plastic inserts.
Design a.—Regular.
Design b.—No bounce.

Type II.—Hammer, solid head.
Class 1.—Copper.
Class 2.—Lead.

Type III.—Mallet.
Class 1.—Rawhide.
Class 2.—Plastic.
Class 3.—Rubber.

Type IV.—Adapter, screw-in head.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

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Federal Specification:

NN–H–93—Handles; Hickory, Striking Tool.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.

Test Method Std. No. 601—Rubber, Sampling, and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL–S–3136—Standard Test Fluids Hydrocarbon and Iso-Octane.

MIL–H–5606 — Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.

MIL–P–15424—Packaging of Hand Tools for Domestic and Overseas Shipment and Storage.

Military Standard:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Governmental:

National Bureau of Standards Handbook:

H28—Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.).

Nongovernmental:

Consolidated Freight Classification Rules and Container Specifications.

(Application for copies should be addressed to Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 General.

3.1.1 *Illustrations.* — The illustrations shown herein are for the convenience of identification and are not intended to preclude the purchase of tools which are otherwise in accordance with the requirements of this specification.

3.2 **Material.** — The material shall be as hereinafter specified for the individual types. The chemical composition and the tempering or curing techniques employed shall be such as to produce tools that conform to this specification.

3.3 **Design.**—The hammers, mallets, and adapters specified herein shall be of the latest design, convenient to use, and entirely adequate for the purpose intended (see 1.1).

3.4 **Identification marking.** — Each hammer, mallet, or adapter shall be permanently and legibly marked on the head, or on the

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handle if made of metal, with the manufacturer's name or trade-mark of such known character that the source of manufacture may be readily determined.

3.5 Finish.—Hammers and mallets except handles shall have either a ground, machined, natural, or enamel finish. Adapters shall have a ground or machined finish.

3.5.1 *Ground or machined finish.*—Ground or machined finish shall be a bright and evenly finished surface, free from deep scratches or excessive roughness.

3.5.2 *Natural finish.*—Natural finish shall be an even, smooth, cast, molded, extruded, rolled, or forged surface, free from scale, rust, or other harmful imperfections.

3.5.3 *Enameled finish.* — Enameled finish shall be a waterproof paint or enamel applied to a surface. The coating shall not be tacky nor flake off readily.

3.6 Weight.—Weight of type I hammers shall include holder (head and handle) and inserts. Weight of type II hammers and type III mallets shall include head and handle.

3.7 Rubber for heads and inserts.—Rubber for mallet heads and rubber inserts for hammers shall conform to table I.

TABLE I.—*Physical properties of rubber*

Tensile strength, p.s.i.	2500 (min.)
Ultimate elongation, percent	200 (min.)
Shore A hardness, instantaneous	95 ± 5
Bashore resilience, 21st impact, percent	15 (max.)
Graves tear strength, pounds per inch, thickness	200 (min.)
Cut resistance, pounds per inch, width	2000 (min.)
Impact indentation, inch:	
At 82° F.	0.04 (max.)
At 120° F.	0.04 (max.)
Volume swell after 48 hours in ASTM oil No. 3:	
At 158° F. percent	10 (max.)

3.8 Eyes.

3.8.1 *Tools with wood handles.*—The head (except for plastic and rubber mallets) shall be provided with an eye which shall extend through the head and in which the handle shall be fitted. Eyes shall have an inward taper at an angle of approximately 3 degrees to permit proper wedging.

3.8.2 *Eye numbers.* — The eye number specified for individual types shall be in substantial agreement with the respective eye number shown on figure 1. (Eye numbers do not apply to tools with steel handles.)

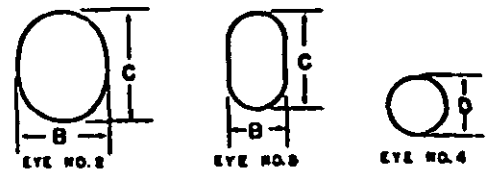


FIGURE 1.—Eye numbers.

3.9 Handles.

3.9.1 *Wood handles.* — Each hammer or mallet shall be fitted with a well-proportioned wood handle, unless hammers with steel handles are specified. Unless grade A is specified, the wood handles shall conform to grade AA of Federal Specification NN–H–93 for the type and size of head specified (see 6.2).

3.9.2 *Steel handles.*—Type II, class 2 lead hammers shall be furnished with steel handles only. Type II, class 1 copper hammers shall be furnished with steel handles when specified in the contract or order. All steel handles shall be of forged steel or seamless or welded steel tubing, well-proportioned and comfortable to the hand. The grip shall be deeply marked or knurled for safe usage. The head shall be molded securely around the handle, pinned, or otherwise secured in such a manner that the handle will not bend or loosen under any working condition. There shall be no openings between the handle and the head where the eye for the wood-handled tool is normally located.

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3.10 Wedges.—Except as specified for the individual types, the eye section of the wood handle shall be secured with one or more metal wedges or metal screws, without or in combination with wood wedges. Wedges and screws shall be driven or screwed, respectively, flush with the front surface of the tool eye. The wedge, wedges, or screws, or a combination of the same, shall create sufficient expansion to completely fill the tool eye and tightly wedge the handle in the eye of the tool, so as to prevent the head from working loose under the most severe working conditions. The metal wedges shall be of the ribbed or corrugated type. The ribs or corrugation shall run in a horizontal or diagonal direction. The vertical direction is represented by the direction in which the wedge is to be driven. The small end of the wedge shall have a slight flat to prevent splitting action of the tool handle. Screws shall be of the tapered, headless type.

3.11 Type I, hammer, inserted face.—Type I hammers may be procured completely assembled or individual inserts and holders may be procured as replacement items (see 6.2).

3.11.1 *Class 1, split head.*—Class 1 hammers shall consist essentially of two inserts of copper, rawhide, lead, plastic, or rubber, as specified (see 6.2), and a holder (head and handle). The inserts shall be cylindrical in shape with the striking surface of the face either flat or slightly convex.

3.11.1.1 *Style A, copper inserts.*—The inserts shall have a copper content of not less than 99 percent and conform to the applicable diameter of faces specified herein.

3.11.1.1.1 *Holder (head and handle).*—The head shall be of malleable iron or steel of two-piece construction held together by a threaded connector and a tightening nut. The head shall be fitted with a hickory handle conforming to 3.9.1. The handles shall be secured with wedges conforming to 3.10 except the wedges may be of wood. In addition, a steel pin shall be inserted transversely through the handle and the fixed section of the head and secured by riveting. The pin shall be located adjacent to the threaded portion of the fixed section. The head shall be so constructed that the inserts may be readily removed, replaced, and interchanged. The inserts shall hold securely under all working conditions.

3.11.1.1.2 Type I, class 1, style A hammers shall be similar to figure 2 and shall conform to table II for the weight specified (see 6.2).

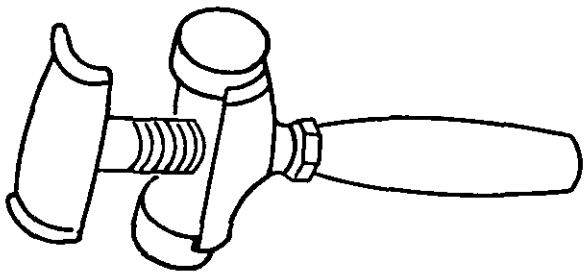


FIGURE 2.—Type I, class 1, split head inserted face hammer.

TABLE II.—*Type I, class 1, style A hammers, inserted face, split head, copper inserts*

Weight [1] +15 percent −5 percent	Diameter of insert faces ± 1/16	Length of head including inserts + 3/4 −1/4	Handle length ± 1	Eye
Pounds	*Inches*	*Inches*	*Inches*	*Number*
2	1 1/4	3 3/8	11	2, 3, or 4
2 3/4	1 1/2	3 5/8	12	2, 3, or 4
4	1 3/4	4	12	2, 3, or 4
6 1/2	2	5 1/8	13	2, 3, or 4

[1] Weight includes head, inserts, and handle.

3.11.1.2 *Style B, rawhide inserts.*—Inserts shall be made of one piece of rawhide, with not more than two filler strips. Hide shall be wound in a smooth spiral, cylindrical in shape. Joints at the end of the spiral shall be smooth and may be formed by a skived and glued lap, or a molded and pinned lap. If pin is used, it shall be not more than 5/16 inch off the

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center of the insert. The striking face shall show no voids and shall be parallel within 1/32 inch. It shall be not more than 2 degrees off the perpendicular, measured at any position on the side of the insert. The edge of the face shall be radiused or chamfered. The sides shall be smooth and free from air pockets, except that cracks or wrinkles not over 1/16 inch deep and at least 1/4 inch from the striking face will be permitted. The inserts shall be completely covered with a moisture-resistant coating. The inserts shall conform to the applicable diameter of insert faces specified herein.

3.11.1.2.1 *Holder (head and handle).*—The holder shall conform to 3.11.1.1.1.

3.11.1:2.2 Type I, class 1, style B hammers shall be similar to figure 2 and shall conform to table III for the weight specified (see 6.2).

TABLE III.—*Type I, class 1, style B hammers, inserted faces, split head, rawhide inserts*

Weight [1] +15 percent −5 percent	Diameter of insert faces ± 1/16	Length of head including inserts + 3/4 − 1/4	Handle length ± 1	Eye
Pounds	*Inches*	*Inches*	*Inches*	*Number*
1 1/4	1 1/4	3 3/8	11	2, 3, or 4
1 3/4	1 1/2	3 5/8	12	2, 3, or 4
2 3/4	1 3/4	4	12	2, 3, or 4
4 1/2	2	5 1/8	13	2, 3, or 4
6	2 3/4	5 7/8	14	2, 3, or 4

[1] Weight includes holder, inserts, and handle.

3.11.1.3 *Style C, lead inserts.*—Inserts shall be of antimonial lead having a Brinell hardness between 10 and 16 and shall be suitable for hammering purposes. The inserts shall conform to the applicable diameter of insert faces specified herein.

3.11.1.3.1 *Holder (head and handle).*—The holder shall conform to 3.11.1.1.1.

3.11.1.3.2 Type I, class 1, style C hammers shall be similar to figure 2 and shall conform to table IV for the weight specified (see 6.2).

TABLE IV.—*Type I, class 1, style C hammers, inserted faces, split head, lead inserts*

Weight [1] +15 percent −5 percent	Diameter of insert faces ± 1/16	Length of head including inserts + 3/4 − 1/4	Handle length ± 1	Eye
Pounds	*Inches*	*Inches*	*Inches*	*Number*
2	1 1/4	3 3/8	11	2, 3, or 4
3	1 1/2	3 5/8	12	2, 3, or 4
4 1/2	1 3/4	4 1/4	12	2, 3, or 4
7	2	5 1/8	13	2, 3, or 4

[1] Weight includes head, inserts, and handle.

3.11.1.4 *Style D, plastic inserts.*—The inserts shall be of plastic or similar material and shall be molded around a metal or hickory disc to provide the necessary rigidity to securely hold the insert under maximum working conditions. The inserts shall have a durometer hardness range between 73 Shore A, 30 second, and 76 Shore D, 30 second. The material shall withstand all test requirements of 4.4.2 except the impact test at minus 30 degrees (see 4.4.2.5) and the permanent distortion test (see 4.4.2.9).

3.11.1.4.1 *Holder (head and handle).*—The holder shall conform to 3.11.1.1.1.

3.11.1.4.2 Type I, class 1, style D hammers shall be similar to figure 2 and shall conform to table V for the weight specified (see 6.2).

3.11.1.5 *Style E, rubber inserts.*—The rubber insert shall have a metal disc or washer vulcanized to one end to provide the necessary rigidity to securely hold the insert under maximum working conditions. The rubber shall conform to the physical requirements of 3.7 and the test requirements under 4.4.1.

3.11.1.5.1 *Holder (head and handle).*—The holder shall conform to 3.11.1.1.1.

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TABLE V.—*Type I, class 1, style D hammers, inserted faces, split head, plastic inserts*

Weight [1] +15 percent −5 percent	Diameter of insert faces ± 1/16	Length of head including inserts + 3/4 −1/4	Handle length ± 1	Eye
Pounds	*Inches*	*Inches*	*Inches*	*Number*
1¼	1¼	3⅜	11	2, 3, or 4
1⅝	1½	3⅝	12	2, 3, or 4
2¼	1¾	4	12	2, 3, or 4
4¼	2	5⅛	13	2, 3, or 4

[1] Weight includes head, inserts, and handle.

3.11.1.5.2 Type I, class 1, style E hammers shall be similar to figure 2 and shall conform to table VI for the weight specified (see 6.2).

TABLE VI.—*Type I, class 1, style E hammers, inserted faces, split head, rubber inserts*

Weight [1] +15 percent −5 percent	Diameter of insert faces ± 1/16	Length of head including inserts + 3/4 − 1/4	Handle length ± 1	Eye
Pounds	*Inches*	*Inches*	*Inches*	*Number*
1⅜	1¼	3⅜	11	2, 3, or 4
1¾	1½	3⅝	12	2, 3, or 4
3	1¾	4	12	2, 3, or 4
4½	2	5⅛	13	2, 3, or 4
6½	2¾	5¼	14	2, 3, or 4

[1] Weight includes head, inserts, and handle.

3.11.2 *Class 2, press-on head.* — Class 2 hammers shall consist essentially of two inserts of plastic and a holder (head and handle). The inserts shall be cylindrical in shape with the striking surface of the face either flat or slightly convex.

3.11.2.1 *Style D, plastic inserts.*—The inserts shall be of plastic or similar material and have a durometer hardness range between 85 Shore A, 30 second, and 76 Shore D, 30 second. The material shall withstand all test requirements of 4.4.2 except impact at minus 30 degrees (see 4.4.2.5), and permanent distortion test (see 4.4.2.9).

3.11.2.1.1 *Holder (head and handle).*—The head shall be of aluminum alloy, zinc alloy, malleable iron, or steel. The design shall be such that will hold the inserts securely under all working conditions but will permit ready removal and replacement of damaged inserts. The head shall be of one or two-piece construction. This type is suitable for yielding materials only but the material shall not be softer than 85 on the Shore A scale. Inserts are secured by expanding on a central core while containing the expansion by a retaining ring. The expanding core shall be not less than half the diameter of the head nor extend into the insert less than the width of the expanding ring. If the head is of two-piece construction, the retaining ring may be separate from the head, but shall be of a material having a tensile strength of not less than 15,000 pounds per square inch. If the head is of one-piece construction, the expanding core shall be contained within a shell, the end of which serves as a retaining ring. The head shall be fitted with a hickory handle conforming to 3.9.1. The handles shall be secured with wedges conforming to 3.10 except the wedges may be of wood.

3.11.2.1.2 *Design a, regular.*—Type I, class 2, style D, design a hammers shall be similar to figure 3 and shall conform to table VII for the weight specified (see 6.2).

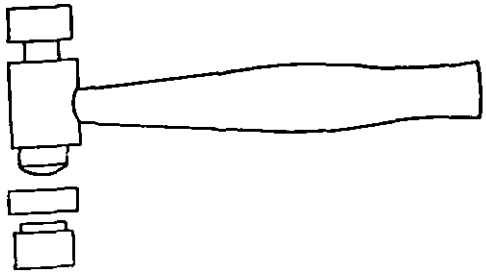


FIGURE 3.—Type I, class 2, style D, design a and b hammers, inserted faces, press-on head, plastic insert faces, regular.

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TABLE VII.—*Type I, class 2, style D, design a hammers, inserted faces, press-on head, plastic insert faces, regular*

Weight [1] +15 percent −5 percent	Diameter of insert faces ± 1/16	Length of head including inserts + 3/4 − 1/4	Handle length ± 1	Eye
Pounds	*Inches*	*Inches*	*Inches*	*Number*
5/8	1 1/4	3 3/4	11 1/2	2 or 3
1	1 1/2	4 1/4	12	2 or 3
1 1/4	1 1/2	4 1/2	12	2 or 3
1 1/2	1 3/4	5 1/2	13 1/4	2 or 3
1 3/4	2 1/4	5 1/2	14 3/4	2 or 3
2 1/4	2 1/4	6	14 3/4	2 or 3
3 1/4	2 1/4	6	14 3/4	2 or 3
3 3/4	2 1/4	6	14 3/4	2 or 3

[1] Weight includes head, inserts, and handle.

3.11.2.1.3 *Design b, no bounce.*—Type I, class 2, style D, design b hammers shall conform to 3.11.2 through 3.11.2.1.1 except the head shall be designed to hold small metal pellets to restrict rebound during normal use but is not required to retain the pellets within the head when the handle is removed. Design b hammers shall be similar to figure 3 and shall conform to table VIII for the weight specified (see 6.2).

TABLE VIII.—*Type I, class 2, style D, design b hammers, inserted faces, press-on head, plastic insert faces, no bounce*

Weight [1] +15 percent −5 percent	Diameter of insert faces ± 1/16	Length of head including inserts + 3/4 − 1/4	Handle length ± 1	Eye
Pounds	*Inches*	*Inches*	*Inches*	*Number*
3/4	1	3 3/4	11	2 or 3
7/8	1 1/4	3 3/4	11 1/2	2 or 3
1	1 1/2	4 1/4	12	2 or 3
1 1/2	1 3/4	5 1/2	13 1/4	2 or 3
2	2 1/4	5 1/2	13 3/4	2 or 3
4	2 1/4	6	13 3/4	2 or 3

[1] Weight includes head, inserts, and handle.

3.11.3 *Class 3, screw-in head.*—Class 3 hammers shall consist essentially of two inserts of plastic and a holder (head and handle). Threads shall be Coarse Thread Series, Class 2B Unified or Class 2 American National in accordance with National Bureau of Standards Handbook H28 and supplement thereto. The dimensions and tolerances shall be such that all inserts and heads of the same nominal diameter can be readily removed, replaced, and interchanged without requiring modification.

3.11.3.1 *Style D, plastic inserts.*—The insert shall be of plastic or similar material and have a threaded stud molded or otherwise permanently fastened into the base of the insert for attachment to the head. Steel studs shall be protected against corrosion by zinc plating. When nonmagnetic studs are specified (see 6.2), the studs shall be of bronze or brass.

3.11.3.1.1 *Hardness of plastic inserts.*—Plastic inserts shall be of seven individual hardness grades with each grade having an identifying color. Plastic inserts shall conform to table IX for the hardness name specified (see 6.2).

TABLE IX.—*Type I, class 3, style D plastic insert hardness grades*

Hardness name	Hardness color	Test instrument and reading	Hardness range
Super soft	Grey	Shore A INSTANTANEOUS	45 to 58
Soft	Brown	Shore A 30 second	45 to 58
Medium	Red	Shore A 30 second	59 to 72
Tough	Green	Shore A 30 second	73 to 85
Medium hard [1]	Cream	Shore D 30 second	40 to 48
Hard [1]	Black	Shore D 30 second	49 to 65
Extra-hard [1]	Yellow	Shore D 30 second	66 to 85

[1] Tests of 4.4.2.5, impact at minus 30 degrees, and 4.4.2.9, permanent distortion, not applicable.
[2] Test of 4.4.2.9, permanent distortion, not applicable.

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3.11.3.1.2 *Shapes and dimensions of plastic inserts.*—Plastic inserts shall be of five individual face shapes and diameters. Unless otherwise specified in the contract or order, the standard face shape shall be furnished. The outside diameter of the standard face shape may be straight or slightly tapered. The insert diameter shall be as specified (see 6.2). The face shapes shall be similar to figures 4A through 4E and the dimensions of the insert identified by the letters A through H of figure 4 shall conform to table X for the face shape and insert diameter specified (see 6.2).

3.11.3.1.3 *Holder (head and handle).*—The head shall be of round cross-sectional shape, shall be of one-piece construction of aluminum alloy or zinc alloy, nonsparking and nonmagnetic, except that the 8- and 10¾-pound hammers may be of ferrous metal. Each end shall have a tapered recess and shall be drilled and tapped to receive the plastic insert. The design shall be such that will insure tightness of the inserts under all working conditions but shall allow the inserts to be readily removed, replaced, and interchanged. The head shall be fitted with a hickory handle conforming to 3.9.1. The handles shall be secured with wedges conforming to 3.10 except the metal wedges shall be of a nonmagnetic material such as zinc alloy, when specified (see 6.2).

3.11.3.1.3.1 *Dimensions of head of holder (without eye).*—The dimensions of the head identified by the letters A through G of figure 5 shall conform to table XI for the diameter specified (see 6.2).

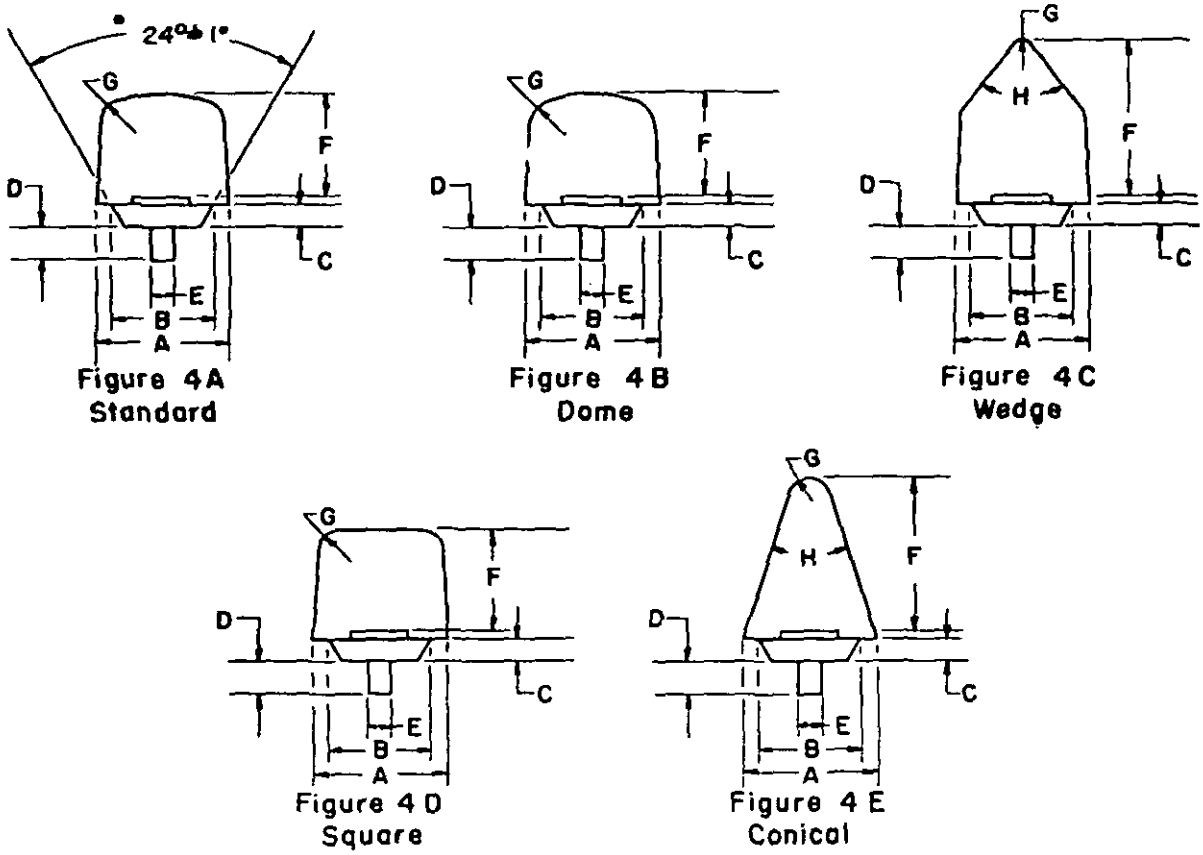


Figure 4A Standard

Figure 4B Dome

Figure 4C Wedge

Figure 4D Square

Figure 4E Conical

*THIS ANGLE IS TYPICAL FOR ALL SIZES AND SHAPES OF INSERTS.

FIGURE 4.—Type I, class 3, style D plastic inserts.

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TABLE X.—*Dimensions of type I, class 3, style D plastic inserts*

Figure number and face shape	A Diameter ± 1/16	B Seat diameter (see note at end of table)	C Seat depth + .030 − .000	D Length of stud ± 1/16	E Diameter of threads nominal	F Usable face ± 1/16	G Radius ± 1/64	H Angle ± 1
	Inches	*Inches*	*Inch*	*Inch*	*Inch*	*Inches*	*Inch*	*Degrees*
4A Standard	1	0.875	0.187	3/16	1/4	0.615	1/32	--
4B Dome	1	.875	.187	3/16	1/4	.615	3/16	--
4C Wedge	1	.875	.187	3/16	1/4	.735	1/8	76
4D Square	1	.875	.187	3/16	1/4	.735	3/32	--
4E Conical	1	.875	.187	3/16	1/4	.840	1/16	23
4A Standard	1 1/2	1.250	.250	3/16	5/16	.650	1/32	--
4B Dome	1 1/2	1.250	.250	3/16	5/16	.650	1/4	--
4C Wedge	1 1/2	1.250	.250	3/16	5/16	1.225	1/8	76
4D Square	1 1/2	1.250	.250	3/16	5/16	1.225	7/64	--
4E Conical	1 1/2	1.250	.250	3/16	5/16	1.275	1/16	23
4A Standard	2	1.750	.312	3/16	3/8	.970	1/16	--
4B Dome	2	1.750	.312	3/16	3/8	.970	3/16	--
4C Wedge	2	1.750	.312	3/16	3/8	1.500	3/16	76
4D Square	2	1.750	.312	3/16	3/8	1.500	5/32	--
4E Conical	2	1.750	.312	3/16	3/8	1.500	3/32	34
4A Standard	2 1/2	2.000	.375	5/16	7/16	1.125	3/32	--
4B Dome	2 1/2	2.000	.375	5/16	7/16	1.125	7/64	--
4C Wedge	2 1/2	2.000	.375	5/16	7/16	1.750	3/16	76
4D Square	2 1/2	2.000	.375	5/16	7/16	1.750	5/32	--
4E Conical	2 1/2	2.000	.375	5/16	7/16	1.750	3/32	34
4A Standard	3	2.500	.500	5/16	1/2	1.250	3/16	--
4B Dome	3	2.500	.500	5/16	1/2	1.250	1/8	--
4C Wedge	3	2.500	.500	5/16	1/2	2.000	1/4	76
4D Square	3	2.500	.500	5/16	1/2	2.000	3/32	--
4E Conical	3	2.500	.500	5/16	1/2	2.000	1/8	34

NOTE.—Tolerance for seat diameters shall be as follows:
Plus 0.025, minus 0.000 for tough and softer grades;
Plus 0.015, minus 0.000 for medium hard and harder grades.

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TABLE XI.—*Dimensions of type I, class 3, style D heads*

A Head diameter ± 0.020	B Largest diameter of seat + 0.000 − .015	C Seat depth + 0.000 − .015	D Included angle ± 1	E Thread diameter Nominal	F Depth of full threads + 0.125 − .000	G Depth of hole (approx.)
Inches	*Inches*	*Inch*	*Degrees*	*Inch*	*Inch*	*Inches*
1	0.875	0.187	30	¼	7/16	5/8
1½	1.250	.250	30	5/16	7/16	11/16
2	1.750	.312	30	3/8	7/16	¾
2½	2.000	.375	30	7/16	5/8	13/16
3	2.500	.500	30	½	5/8	1 1/16

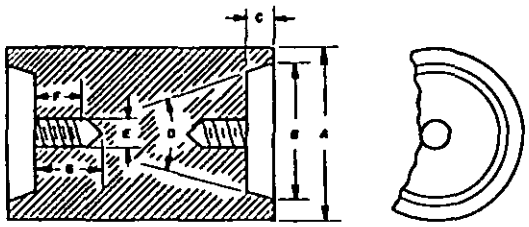


FIGURE 5.—Identification of dimensions for type I, class 3, style D head (without eye).

3.11.3.1.4 *Design a, regular.*—Type I, class 3, style D, design a, hammers shall be similar to figure 6 and conform to table XII for the weight, insert diameter, and length of head including standard faces specified (see 6.2).

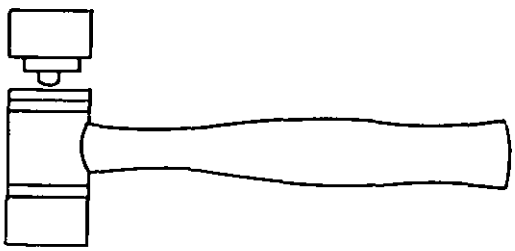


FIGURE 6.—Type I, class 3, style D, designs a and b hammers, inserted face, screw-in head, regular and no bounce, plastic inserts.

3.11.3.1.5 *Design b, no bounce.*—Type I, class 3, style D, design b hammers shall conform to 3.11.3 through 3.11.3.1.3.1 except the head shall be designed to hold small metal pellets to restrict rebound during normal use. The head shall be of such design that the pellets will be retained within the head when the handle is removed or, being inserted. Design b hammers shall be similar to figure 6 and shall conform to table XIII for the weight specified (see 6.2).

3.12 Type II hammers, solid head.

3.12.1 *Class 1, copper.*—The head of copper hammers shall be square or barrel-shaped and shall be of not less than 99 percent copper. Square heads shall have chamfered corners on upper and lower portions. Both faces shall be smooth and the remainder of the head shall have a natural or enamel finish. The head shall be fitted with a hickory handle conforming to 3.9.1, or when specified (see 6.2), steel handles conforming to 3.9.2 shall be furnished. Wood handles shall be secured with wedges conforming to 3.10. The shape of the head and the eye number shall be optional with the contractor unless otherwise specified in the contract or order (see 6.2). Eye numbers do not apply to heads with steel handles.

3.12.1.1 Type II, class 1 copper hammers shall be similar to a or b of figure 7 and shall conform to table XIV for the weight specified (see 6.2).



FIGURE 7.—Type II, class 1 copper hammer.

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TABLE XII.—*Type I, class 3, style D, design a hammers, inserted face, screw-in head, plastic inserts, regular*

Weight [1] Tolerances shown in ounces	Diameter insert faces ± 1/8 heads ± 0.020	Length of head including standard inserts ± 3/8	Length of head without inserts + 0.060 − .030	Handle length ± 1	Eye
	Inches	*Inches*	*Inches*	*Inches*	*Number*
6 ounces ± 1	1	3 3/8	2	11	2 or 3
9 ounces ± 1	1	3 3/8	2	11	2 or 3
3/4 pound ± 1 1/2	1 1/2	4 1/4	2 3/8	13	2 or 3
1 1/4 pounds ± 2	1 1/2	6 1/4	4 1/2	13	2 or 3
1 1/4 pounds ± 2	1 1/2	4 1/4	2 3/8	13	2 or 3
1 1/4 pounds ± 2	2	4 3/4	2 1/2	13	2 or 3
2 pounds +3 −2	2	4 3/4	2 1/2	13	2 or 3
3 1/4 pounds +5 −3	2	6 1/8	3 3/8	15	2 or 3
2 1/2 pounds +4 −3	2 1/2	6	3 1/4	15	2 or 3
4 pounds +6 −4	2 1/2	6	3 1/4	13	2 or 3
3 1/2 pounds +5 −3	3	6 1/4	3 3/4	14	2 or 3
6 1/2 pounds +8 −5	3	6 1/4	3 3/4	14	2 or 3
8 pounds +10 −5	3	7 5/8	4 5/8	28	2 or 3
10 3/4 pounds +12 −6	3	8 3/4	5 3/4	36	2 or 3

[1] Weight includes head, inserts, and handle.

TABLE XIII.—*Type I, class 3, style D, design b hammers, inserted face, screw-in head, plastic inserts, no bounce*

Weight [1] Tolerances shown in ounces	Diameter insert faces ± 1/8 heads ± 0.020	Length of head including standard inserts ± 3/8	Length of head without inserts + 0.060 − .030	Handle length ± 1	Eye
	Inches	*Inches*	*Inches*	*Inches*	*Number*
1 pound ± 2	1 1/2	5 1/4	3 1/2	13	2 or 3
1 1/2 pounds +3 −2	1 1/2	6 1/4	4 3/8	13	2 or 3
2 1/2 pounds +4 −3	2	6 3/8	4 3/8	13	2 or 3
3 1/2 pounds +5 −3	2	8 3/4	6 1/2	14	2 or 3
5 1/2 pounds +7 −4	3	8	5	14	2 or 3

[1] Weight includes head, inserts, and handle.

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TABLE XIV.—*Type II, class 1 copper hammers*

Weight [1] Tolerances shown in ounces	Length of head	Diameter of head or distance across flats	Handle length (overall)	Eye		
				Number	Dimensions, minimum	
					B	C
Pounds	*Inches*	*Inches*	*Inches*		*Inch*	*Inches*
1½ ± 2	2¼ ± ¾	⅞ + ⅜ − ⅛	11 ± 1	2 or 3	5/16	9/16
1 ± 2	3 ± ¾	1 + ⅜ − ⅛	12 ± 1	2 or 3	5/16	9/16
1½ ± 3	3 ± ¾	1¼ + ⅜ − ⅛	12 ± 1	2 or 3	5/8	⅞
2 ± 3	3¾ ± ¾	1¼ + ⅜ − ⅛	13 ± 1	2 or 3	5/8	⅞
3 ± 5	4 ± ¾	1½ + ⅜ − ⅛	14 ± 1	2 or 3	¾	1
4 ± 5	4½ ± ¾	1¾ + ⅜ − ⅛	15 + 2 − 1	2 or 3	¾	1
6 ± 5	5 ± ¾	2 + ⅜ − ⅛	18 + 2 − 1	2 or 3	¾	1
8 ± 8	6 ± 1	2 + ⅜ − ⅛	19 + 2 − 1	2 or 3	1	1¼
10 ± 8	7 ± 2	2 + ⅞ − ⅛	20 + 2 − 1	2 or 3	1	1¼
12 ± 8	8 ± 2	2 + ⅞ − ⅛	22 + 2 − 1	2 or 3	1	1¼

[1] Weight includes head and handle.

3.12.2 *Class 2, lead.* — The head of lead hammers shall be antimonial lead having a Brinell hardness of 10 to 16. The head shall be barrel-shaped and shall have a smooth natural finish all over. The head shall have a steel handle conforming to 3.9.2.

3.12.2.1 Type II, class 2 lead hammers shall be similar to figure 8 and shall conform to table XV for the weight specified (see 6.2).



FIGURE 8.—Type II, class 2 lead hammer.

3.13 Type III mallet.

3.13.1 *Class 1, rawhide.*—The head of class 1 mallets shall be made of one piece of rawhide, with not more than two filler strips. Hide shall be wound in a smooth spiral, cylindrical in shape. Joints at the end of the spiral shall be smooth and may be formed by a skived and glued lap, or a molded and pinned lap. The pin shall be not more than 3/16 inch off the center of the head. The striking face shall show no voids, shall be parallel within 1/32 inch and shall be 90 ± 2 degrees to the

TABLE XV.—*Type II, class 2 lead hammers*

Weight[1] Tolerance shown in ounces	Diameter of head ±⅛	Handle length (overall) ±1
Pounds	*Inches*	*Inches*
1 +3 −1	1	12
1½ +3 −2	1¼	12
2½ +4 −2	1¼	13
3 +4 −3	1½	14
3½ +4 −3	1½	14
5 +6 −4	1¾	16
6 +8 −4	2	18

[1] Weight includes head and handle.

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sides of the head. The edge of the face shall be radiused or chamfered. The sides shall be smooth and free from air pockets, except that cracks or wrinkles not over 1/16 inch deep and at least 1/4 inch from the striking face will be permitted. The head shall be completely covered with a moisture-resistant coating. The head shall be fitted with a hickory handle conforming to 3.9.1. The head may have a straight hole and be pinned or otherwise permanently fastened to the handle in lieu of taper hole and wedges.

3.13.1.1 Type III, class 1 rawhide mallets shall be similar to figure 9 and shall conform to table XVI for the weight specified (see 6.2).

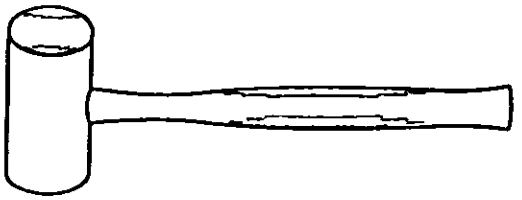


FIGURE 9.—Type III, class 1 rawhide mallet.

TABLE XVI.—*Type III, class 1 rawhide mallets*

Weight[1] (Approx.)	Length of head ±1/4	Diameter of head ±1/8	Handle length (overall) ±1
Ounces	*Inches*	*Inches*	*Inches*
2	2 1/4	1	11
4	2 3/4	1 1/4	12
6	3	1 1/2	12
8	3 1/4	1 3/4	13
11	3 1/2	2	13
22	4 1/2	2 3/4	14
24	4 3/4	2 3/4	14

[1] Weight includes head and handle.

3.13.2 *Class 2, plastic.*—The head of class 2 mallets shall be composed of molded plastic and, except when otherwise specified herein, the head shall withstand the tests specified in 4.4.2 for plastic insert faces. Plastic heads shall be of four individual hardness grades with each grade having an identifying color. The heads shall conform to table XVII for the hardness name specified (see 6.2). The head shall be fitted with a hickory handle conforming to 3.9.1. The head may have a straight hole and shall be pinned or otherwise permanently fastened to the handle in lieu of taper hole and wedges. Both faces shall be straight or slightly convex and the sides shall be straight or slightly tapered toward the faces. The head shall have a smooth natural finish.

3.13.2.1 Type III, class 2 plastic mallets shall be similar to figure 10 and conform to table XVIII for the weight specified (see 6.2).

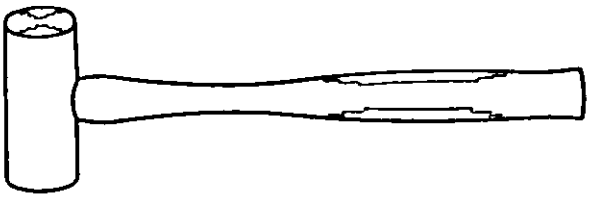


FIGURE 10.—Type III, class 2 plastic mallet.

3.13.3 *Class 3, rubber.*—The head of class 3 mallets shall be barrel-shaped, of solid resilient rubber, suitable for automotive body, fender, and battery work, where marring of the work is to be avoided. Both faces shall be either flat or slightly convex and the head shall have a smooth, natural finish. The head shall have a hickory handle conforming to 3.9.1. The head shall be securely and permanently bonded with the handle in such a manner that there is no separation of rubber

TABLE XVII.—*Type III, class 2 plastic mallet head hardness grades*

Hardness name	Hardness color	Test instrument and reading	Hardness range
Soft	Brown	Shore A - 30 second	45 to 58
Medium	Red	Shore A - 30 second	59 to 72
Tough	Green	Shore A - 30 second	73 to 85
Hard[1]	Black	Shore D - 30 second	49 to 65

[1] Test of 4.4.2.9, permanent distortion, not applicable.

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TABLE XVIII.—*Type III, class 2 plastic mallets*

Weight[1] +15 percent —10 percent	Length of head +⅝ —¼	Diameter of head +⅛ —1/16	Handle length (overall) ±1
Ounces	*Inches*	*Inches*	*Inches*
6	3¼	1½	11
8	3¼	1¾	12
12	3½	2	12
24	4¼	2⅝	13

[1] Weight includes head and handle.

from the handle under the most severe working conditions. The rubber head shall conform to the physical requirements of 3.7 and shall withstand the tests specified in 4.4.1.

3.13.3.1 Type III, class 3 rubber mallets shall be similar to figure 11 and shall conform to table XIX.

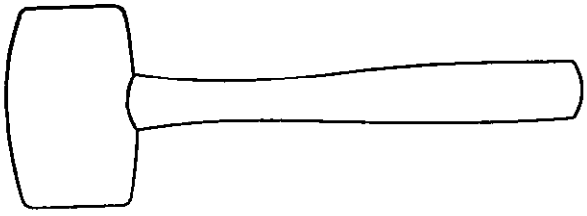


FIGURE 11.—Type III, class 3 rubber mallet.

TABLE XIX.—*Type III, class 3 rubber mallet*

Weight[1] ±3	Length of head ±⅝	Diameter of head +¾ —¼	Handle length (overall) ±1
Ounces	*Inches*	*Inches*	*Inches*
24	4	2	15

[1] Weight includes head and handle.

3.14 **Type IV, adapter, screw-in head.**—Adapters shall be of high grade steel and fully heat-treated to be suitable for use with portable power tools. One end shall have a tapered recess and shall be drilled and tapped to receive either a 1½- or 2-inch nominal diameter plastic insert. The design shall be such that will insure tightness of the insert under all working conditions but shall allow the insert to be readily removed, replaced, and interchanged. The adapters shall have a ground or machined finish and shall be protected against corrosion by zinc plating.

3.14.1 *Plastic inserts for adapters.*—The 1½- and 2-inch nominal diameter plastic inserts shall conform to 3.11.3.1 through 3.11.3.1.2.

3.14.2 *Dimensions of adapter.*—The dimensions of the adapter identified by the letters A through F and the dimensions shown on figure 12 shall conform to table XX for the adapter shank size and head diameter specified (see 6.2). Dimensions without tolerances shown on the figure are approximate.

3.15 **Workmanship.** — Workmanship shall be first class in every respect. The tools shall have no burrs, fins, sharp projections, or any other imperfections which may impair their durability and serviceability.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Inspection lot.**—All hammers, inserts, adapters, or mallets of the same size, style, class, type and design offered for delivery at one time shall be considered a lot for purposes of acceptance inspection and tests.

4.2 **Sampling for acceptance inspection.**

4.2.1 *Sampling for visual and dimensional examination.* — A random sample of hammers, adapters or mallets shall be selected from each inspection lot of material offered for Government inspection of visual and dimensional characteristics in accordance with Military Standard MIL–STD–105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300, and inspection level I for lots over 300. The acceptable quality level shall be equal to 1.5 percent defective.

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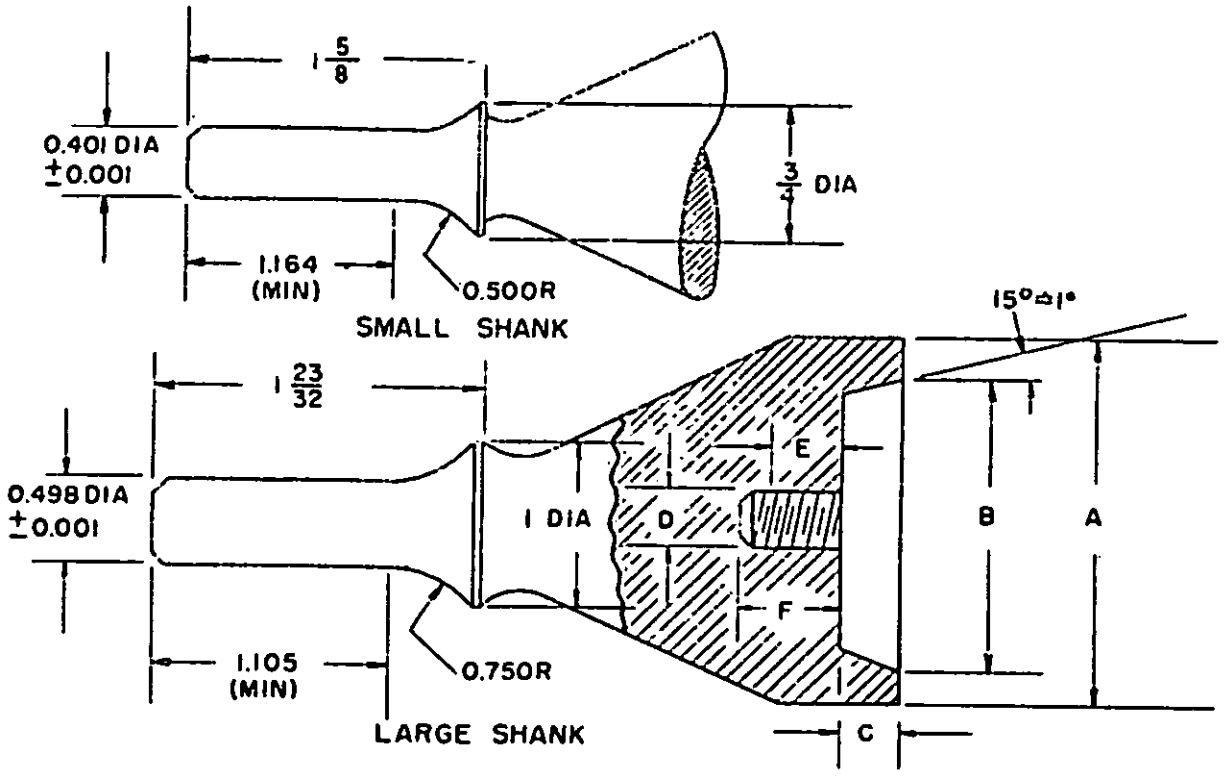


FIGURE 12.—Type IV adapter, screw-in head.

TABLE XX.—*Dimensions of type IV adapters, screw-in head*

Shank size ±0.001	A Head diameter ±0.020	B Largest diameter of seat +0.000 — .015	C Seat depth +0.000 — .015	D Thread diameter NC-2 nominal	E Depth of full threads +0.125 — .000	F Depth of hole (approx.)
Inch	*Inches*	*Inches*	*Inch*	*Inch*	*Inch*	*Inch*
0.401	1⅛	1.250	0.250	5/16	1/16	11/16
.401	2	1.750	.312	⅜	1/16	¾
.498	1⅛	1.250	.250	5/16	1/16	11/16
.498	2	1.750	.312	⅜	1/16	¾

4.2.2 *Sampling for tests of rubber mallets and rubber inserts.*—The manufacturer shall provide a sheet of rubber material 0.07 to 0.08 inch thick, having a total surface area of at least one square foot, together with at least eight solid heads or insert-face blanks which shall be of the same composition and cure, and prepared at the same time from material used in the heads or inserts furnished. Each sample shall be tested as specified in 4.4.1 to verify compliance with the physical properties specified in 3.7.

4.2.3 *Sampling for tests of plastic inserts.*—The manufacturer shall provide at least eight solid heads or insert face blanks of the plastic material which shall be of the same composition and cure, and prepared at the

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same time as the material used in the heads or inserts furnished. Each sample shall be tested as specified under 4.4.2 to verify compliance with this specification.

4.2.4 *Sampling for striking test (see 4.4.2.6).* — A random sample of hammers having plastic insert faces and plastic mallets shall be selected from each inspection lot in accordance with the appendix to Military Standard MIL–STD–105 at inspection level L–7. The acceptable quality level shall be equal to 2.5 percent defective.

4.3 Visual and dimensional examination.— Each of the sample hammers, adapters, or mallets selected in accordance with 4.2.1 shall be visually and dimensionally inspected by the inspector to verify compliance with this specification. Any hammer, adapter, or mallet in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective hammers, adapters, or mallets, in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection provided the contractor has removed all nonconforming hammers, adapters, or mallets. Samples shall again be selected from such rejected lots and examined to verify compliance with this specification.

4.4 Test procedures.

4.4.1 *Tests for rubber mallet heads and inserts.*—Rubber for mallet heads and inserts shall be tested in accordance with 4.4.1.1 through 4.4.1.7.

4.4.1.1 Tensile strength and ultimate elongation shall be determined on specimens by the procedure described in Federal Test Method Standard No. 601.

4.4.1.2 Hardness measurements shall be made on the mallet head or insert face with the Shore type A instrument. The average of measurements on duplicate specimens shall be reported.

4.4.1.3 *Resilience test.* — Test specimens shall be disks ½ inch thick by at least 1¼-inch diameter and shall be buffed flat on both sides. Insert-face blanks of the same width may be used.

4.4.1.3.1 *Apparatus.*—The sample shall be positioned so that the point of impact on the plunger shall be at the point of maximum thickness of the section of sample to be tested. The sample shall be held in position by the weight of the adjustable unit of resiliometer. The specimen shall be given a warm up to 20 impacts at approximately 2 impacts per three seconds. The average of the next 10 readings shall be taken as the resilience of the specimen. Testing shall be done at 82° F. (28° C.) ± 5° F. after the sample shall have been conditioned for at least 30 minutes at this temperature.

4.4.1.4 *Determination of tear resistance.*

4.4.1.4.1 *Test specimen.* — The test specimens shall be cut from sheets approximately 0.08 inch thick by 6 inches square by means of a Graves angle die.

4.4.1.4.2 *Apparatus.*—The apparatus used for the determination of tear resistance shall be a standard Scott rubber tester.

4.4.1.4.3 *Procedure.* — The specimen shall be clamped in the jaws of the testing machine so that the longitudinal axis is in line with the direction of the application of the load. The load shall be applied uniformly. Resistance shall be expressed in pounds pull per inch thickness.

4.4.1.5 *Swelling test.*

4.4.1.5.1 *Test specimens.*—The test specimens shall be rectangular synthetic rubber blocks 2 by 1 inch and not over 1/16 inch thick. Three specimens shall be used for each test and the results averaged.

4.4.1.5.2 *Procedure.*—The volume of each test specimen shall be measured by the water

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displacement method, in which the specimen is accurately weighed to the nearest mg in air (W1) and in distilled water (W2) at room temperature. When weighing in water, care shall be taken that the specimen is free from adhering air bubbles and, if necessary, it may first be wetted by dipping into 95 percent ethyl alcohol and thoroughly rinsed with distilled water. After weighing, the specimen shall be blotted dry with filter paper, completely immersed in ASTM oil No. 3 in a flask, and allowed to stand for 46 ± ¼ hours at 158° F. Excessive evaporation shall be avoided by fitting the container with a cork stopper. At the termination of the immersion period, the specimen shall be removed from the specinl testing fuel, dipped in 95 percent alcohol, blotted lightly with filter paper, and placed in a tared weighing bottle and weighed (W3). The specimen shall then be removed from the bottle and weighed in distilled water (W4) in immediate consecutive procedure to determine the displacement after test. The final weighing shall be completed within 3 minutes after removal from the special testing fuel.

4.4.1.5.2.1 *Calculation.* — The increase in volume shall be calculated as follows:

$$\text{Percentage increase in volume} = \frac{(W3 - W4) - (W1 - W2) \times 100}{W1 - W2}$$

4.4.1.6 *Cut resistance.* — Test specimens shall be ½ by 1 by 1 inch which have been dusted lightly with talc. Insert-face blanks may be used after being cut to this size.

4.4.1.6.1 *Apparatus.*—The cutter shall be the same as that normally supplied with the Bell Telephone compression cutting tester, the cutter to be wedge shaped having an included angle of 22 degrees. At the apex of the wedge, a steel music wire 0.046 inch in diameter shall be imbedded.

4.4.1.6.2 *Procedure.* — The specimen shall be placed on the anvil and the cutter brought down against the specimen at a rate of 2 inches per minute. Cutting will be evidenced by a sharp decrease in indicated load.

4.4.1.7 *Impact resistance test.*—Impact resistance is defined in this case as the resistance to permanent indentation by a free falling weight weighing 10 pounds with a hemispherical indentor ½ inch in diameter on its striking surface, the weight falling from a height of 5 feet. The weight shall be cylindrical in shape and shall be dropped through a piece of seamless tubing slightly larger than the diameter of the weight in order to insure that the full force of the falling weight will be acting normal to the striking surface. Test specimens shall be ½ inch thick by at least 1¼-inch diameter and placed upon a heavy steel plate or similar foundation. Insert-face blanks may be used. The rubber material shall show no sign of chipping or cracking after the impact test.

4.4.1.8 The rubber insert or head shall be free from patches, cracks, holes, blisters, decontaminations, soft spots, or any other defect not specifically permitted by the contract or order.

4.4.2 *Tests for plastic insert faces and mallet heads.*

4.4.2.1 Plastic for insert faces and mallet heads shall be tested in accordance with the tests specified in 4.4.2.2 through 4.4.2.9. A new specimen shall be used for each test.

4.4.2.2 *Flammability test.* — The sample shall be clamped in a support in a horizontal position. Under the sample there shall be clamped a piece of 10-mesh Bunsen-burner gauze in a horizontal position, ¼ inch below the bottom of the sample with approximately ½ inch of the sample extending beyond the edge of the gauze. A Bunsen burner or alcohol lamp, with a flame of ½ inch to ¾ inch in height shall be placed under the free end of the sample and adjusted so that the flame tip is just in contact with the sample. At the end of 30 seconds the flame shall be removed and the plastic allowed to burn. The rate of

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travel of the flame shall be determined from the time required for the flame to travel the length of the sample. Should burning cease after the first ignition, the Bunsen burners shall be placed under the free end immediately following the extinction of the flame for a second period of 30 seconds. If the sample does not continue to burn it shall be considered as self-extinguishing. If the sample continues to burn, the burning rate shall not exceed 2 inches per minute.

4.4.2.3 *Heat distortion.*—The sample for test shall be suspended in boiling water at 212° F. for a period of 2 minutes. At the end of this period it shall be removed from the water and shall show no significant distortion or blistering. Significant distortion or blistering (for purpose of this test) means an increase or decrease of at least 4 percent in the original dimensions of the sample, either of a uniform or irregular bulge, blister, or depression.

4.4.2.4 *Impact test at room temperature.*—Samples shall be placed, screwed or otherwise arranged to simulate actual use on a steel plate. A cylindrical weight for the specific size, shall be dropped through a length of seamless tubing, slightly larger than the weight, to insure full force of the weight acting on the normal striking face. The sample shall not break, crack or significantly distort when the weight has been dropped a minimum of 20 times from a height determined by table XXI.

TABLE XXI.—*Impact test at room temperature*

Diameter of insert	Weight	Drop
Inches	*Pounds*	*Inches*
1 to 1¼	10	28
1½ to 1¾	10	33
2 to 2¼	15	38
2½ to 2⅝	15	45
3	25	42

4.4.2.5 *Impact at minus 30 degrees.*—Sample hammer insert or mallet head (except medium hard) shall be kept at a temperature of minus 30° ± 5° F. for 5 hours and subjected to the tests described in 4.4.2.4 except that only one blow shall be struck. Weight and drop shall be in accordance with table XXII. The tests shall be completed within 1 minute after removal of the sample from the cooling medium or cabinet. Failure of more than 10 percent of the units tested will constitute rejection. A sample is deemed to have failed if it breaks or cracks.

TABLE XXII.—*Impact test at 30 degrees*

Face diameter	Weight	Drop
Inches	*Pounds*	*Inches*
1	10	11
1½ to 1¾	10	12
2	15	15
2½ to 2⅝	15	18
3	15	28

4.4.2.6 *Striking test.*—Each sample hammer or mallet selected in accordance with 4.2.4 shall withstand 25 full swinging blows on the outside corner of heavy 2 by 2 angle iron or block of steel. The blows should be of the maximum force which a representative workman of the trade is capable of delivering through repeated hand striking. The corner shall be smoothed to a radius of not less than 1/32 nor more than 3/64-inch. The angle iron or steel block shall be placed on an anvil, concrete or other solid base, to assure maximum delivered blow to the tool being tested. The insert or head shall have failed if there are any permanent impressions, displacement of face material, flaking, mushrooming, or any splits or gouges over 1/32 inch deep, running more than 50 percent of the face diameter.

4.4.2.7 *Solvent test.*—Test specimen shall be completely submerged in hydraulic oil in accordance with Military Specification MIL–H–5606 and hydrocarbon fluid in accordance with Military Specification MIL–S–3136 for a period of 24 hours in each solution. Specimen shall show no greater than 2 percent increase

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or 1 percent decrease in volume after removal from the solvent.

4.4.2.8 *Hardness test (durometer).*—Samples shall be soaked for a minimum of 2 hours at 68° ± 2° F. and the hardness tested within 1 minute after removal from cabinet. Using a durometer A–2 or D–2 instrument, as applicable, the specimen shall be placed on the instrument and pressure applied, the hardness reading shall be taken 30 seconds after contact, except that for the super soft grade an instantaneous reading shall be taken. Durometer hardness shall fall within tolerances shown in 3.11.1.4, 3.11.2.1, table IX or XVII as applicable.

4.4.2.9 *Permanent distortion test.* — Samples shall be soaked for 2 hours at 68° ± 2° F. and tested within 2 minutes as follows: Samples shall be clamped in a vise against the threaded surface of a national course thread series ½-inch bolt. The maximum flat or radius at the crest of the threads shall not exceed 0.010 inch. The vise shall be tightened until the bolt imbeds itself ¼ inch into the head or insert face and after being held in this position for a period of 1 minute, the vise shall be loosened and the specimen removed. After a period of not less than 30 seconds or longer than 10 minutes the specimen shall not evidence any permanent marring or distortion and will have returned to its original shape. Material returning to shape in less than 30 seconds or which requires more than 10 minutes to recover shall be considered unsatisfactory.

4.4.2.10 Samples shall be free from patches, cracks, holes or any other defects, not specifically permitted by the contractor or order, that may effect their serviceability.

4.4.2.11 Plastic inserts shall be considered to have failed if after heat distortion, solvent or striking tests, the insert cannot be properly attached to the body as originally installed.

5. PREPARATION FOR DELIVERY

(Nonmilitary agencies.—Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.*—Level A cleaning, preservation, and packaging shall be in accordance with level A in Military Specification MIL–P–15424.

5.1.2 *Level C.*—Level C cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Level A packing shall be in accordance with level A in Military Specification MIL–P–15424.

5.2.2 *Level B.*—Level B packing shall be in accordance with level B in Military Specification MIL–P–15424.

5.2.3 *Level C.*—Hand tools, preserved and packaged in accordance with level A or C as specified (see 6.2), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the Consolidated Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Military agencies.*—Marking shall be in accordance with Military Specification MIL–P–15424.

5.3.2 *Nonmilitary agencies.*

5.3.2.1 *Interior containers.*—Packages and interior containers shall be marked with the name of the material and the name or trademark of the contractor, of such known char-

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acter that the source of supply may be readily determined.

5.3.2.2 *Exterior containers.* — Shipping containers shall be marked with the name of the item, the type and class, as defined by the contract or order under which shipment is made, the quantity contained therein, the name of the contractor, and the number of the contract or order.

6. NOTES

6.1 Intended use. — Inserts for hammers and adapters, and solid head hammers and mallets are designed to prevent marring or damaging of work of many varieties. Since many factors are involved it is considered impractical to designate specific tools for specific work. Therefore, the following is offered as a guide in selecting the most suitable tool for the work:

(*a*) Maximum life of the tool may be obtained by using the softest grade of insert tip or mallet that will accomplish the work.

(*b*) Watch for these signs:

(1) When tool marks the work, select the next grade softer.

(2) When the tool face wears excessively, select the next grade softer or select a similar weight tool in the next series larger diameter.

(3) When the tool face does not drive or accomplish the work, select the next grade harder.

(4) For hammers with replaceable inserts, consider using two different hardness inserts as required.

(*c*) Proper tools may be selected for work requiring use of nonsparking or nonmagnetic tools.

6.2 Ordering data. — Procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, style, design, and weight (when applicable) (see 1.2 and applicable table).

(*c*) For hammers or mallets with wood handles, specify when grade A wood handles are acceptable (see 3.9.1).

(*d*) For type I, class 3, style D, design a, specify insert diameter (see 3.11.3.-1.2); specify length of head including standard inserts (see table XII); specify hardness name of plastic insert (see 3.11.3.1.1); specify when nonmagnetic studs are required (see 3.11.3.1); specify shape when other than standard shape insert is required (see 3.11.3.1.2); specify when nonmagnetic metal wedges for the handle are required (see 3.11.3.1.3).

(*e*) For type II, class 1, specify when steel handle is required (see 3.12.1); specify when particular shape of head is required (see 3.12.1); specify when particular eye number is required (see 3.12.1).

(*f*) For type III, class 2 specify hardness name of plastic head (see 3.13.2).

(*g*) For type IV adapters, specify shank and head diameter (see 3.14.2).

(*h*) When only inserts are required, specify type, class, style, and diameter; for type I, class 3, style D, and type IV inserts specify diameter, hardness name, and shape (if other than standard shape is required).

(*i*) When only holders are required, specify the type, class, weight (of the hammer) and insert face diameter; for type I, class 3, style D holders, specify weight (of the hammer), insert face diameter, and length of head without inserts (see table XII).

(*j*) Whether preservation, packaging, and packing shall be for level A, B or C (see 5.1 and 5.2).

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6.3 Superseding data.—Types, classes, and styles of soft faced hammers and mallets covered in this specification supersede those covered in Federal Specification GGG–H–86a, as follows:

This specification	*Federal Specification GGG–H–86a*
Type I, class 1, style A	Type K, class I
Type I, class 1, style B	Type K, class II
Type I, class 1, style C	Type K, class III
Type I, classes 1, 2, and 3, style D	Type K, class IV
Type I, class 1, style E	Type K, class V
Type I, classes 2 and 3, style D, design b	Type K, class VI
Type II, class 1	Type E
Type II, class 2	Type X
Type III, class 2	Not covered
Type III, class 3	Type O
Type IV	Not covered

6.3.1 Type III, class 1 of this specification supersedes type VII of Federal Specification LLL–M–71a.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTEREST:
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